

DETAILED ASSIGNMENT GUIDELINES

Exploring Intelligent Systems: From Independent Study to Practical Implementation

1. Objective

This assignment aims to help students develop self-learning skills, conduct independent research, and broaden their knowledge in areas of Information Systems / Artificial Intelligence beyond the core curriculum. Students are expected to enhance their critical thinking and ability to apply theoretical knowledge to real-world problems.

2. Topic List

Each student is free to choose topics for self-study from the following list:

- Association Rules Mining
- Recommender Systems
- Fuzzy Logic
- Linear Regression
- Logistic Regression
- Latent Dirichlet Allocation (LDA)
- Deep Neural Networks
- Word Embedding

3. Detailed Requirements

3.1. Essay

Each student selects **two topics** from the list above and studies them over a period of 4 weeks. For each topic, students must write one essay (Total: 2 essays).

- **Suggested structure:** Problem introduction, Core concepts, Mathematical/logical foundations, Practical applications, Advantages and disadvantages.
- **Requirement:** Minimum 1,000 words per essay, clearly presented, with illustrations if necessary.

3.2. Lab Assignment

Each student must design and implement **one lab assignment** based on at least one of the selected topics.

- **Required components:** Problem Statement, Dataset (source and structure), Source code, and Output explanation.
- **Format:** Report file with source code (Jupyter Notebook or Python script).

3.3. Weekly Learning Journal

This is a mandatory requirement to track progress. At the end of each week, students must submit a short report:

- What did you learn this week?
- What did you understand?
- What difficulties did you encounter?
- Plan for next week

4. Grading Rubric

Component	Evaluation Criteria	Weight	Requirement for Full Score
1. Weekly Reports	Discipline & Consistency	10%	Submit all 4 reports on time.
	Reflection Quality	10%	Clearly describe learning progress.
2. Essays	Knowledge & Content	15%	Accurate explanation of concepts.
	Extension Ability	15%	Provide real-world examples.
	Presentation & References	10%	Proper structure and references.
3. Lab Assignment	Problem Design	10%	Clear and logical problem.
	Implementation	15%	Working code with explanation.
	Result Evaluation	15%	Analyze output results.

5. Registration and Submission Guide

Use this to submit your Weekly Journal and Ask question.

- **Link:** <https://support-learning.vercel.app/register>

Illustration:**Register****Class code *** e.g. IS-1, IS-2, IS-3**Full name** Enter your full name**Username** Enter your username**Password** Enter your password**Confirm password** Re-enter your password **Register**Already have an account? [Login](#)**Class Code Instructions:**

- CC01: "IS-1"
- IM01: "IS-2"
- CH01: "IS-3"

6. Submission Rules & Deadline

- **Weekly:** Submit Weekly Journal.
- **The Last Week of April (3/5)** Submit the final package (2 Essays + 1 Lab).